

# LIU Jiahang

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Research Interests: Human–Robot Interaction, Dexterous Teleoperation, Robotics, Computer Vision, Multimodal Fusion

## EDUCATION

### Dalian University of Technology

B.Eng in Mechanical Design, Manufacturing and Automation

GPA: 3.35 / 4.00 (83.5 / 100) | Award: Social Practice Scholarship

Dalian, China

2020.09 – 2024.07

### Dalian University of Technology

M.S. Student in Mechanical Engineering, School of Mechanical Engineering

Dalian, China

2024.09 – 2027.07

### University of Tokyo

Special Research Student, Mechanical Engineering, Graduate School of Engineering

Tokyo, Japan

2025.10 – 2026.04

## PUBLICATIONS

- Liu J., Li Z., Le R., et al. “Precise Hand-Arm Teleoperation of a Dexterous Robotic Manipulator via Markerless Vision and High-Density sEMG Fusion.” *IEEE Robotics and Automation Letters*, 2025, 11(2), 1938–1945.
- Ying Z., Zhang W., Liu J., et al. “VEMG-TeleGrasp: A Stiffness-Aware Teleoperation Framework for Dexterous Manipulation via Multimodal Fusion.” *IEEE/ASME Transactions on Mechatronics*, 2026. Accepted.

## EXPERIENCE

### Multimodal Hand Joint-Angle Regression from Egocentric Vision and Forearm High-Density sEMG

Dalian University of Technology / University of Tokyo (*under review*)

2025.09 – present

- Developed and evaluated a finger-unencumbered multimodal framework that combines egocentric RGB images and 128-channel forearm HD-sEMG signals for continuous regression of 20 hand joint DoFs.
- Proposed a DoF-query Transformer fusion decoder that uses independent learnable queries to integrate ViT-based visual representations with Transformer-encoded EMG features for DoF-specific and context-adaptive fusion.
- Established a synchronized multimodal acquisition and processing pipeline for 10 participants, integrating head-mounted RGB imaging, 128-channel forearm HD-sEMG, and Vicon optical motion capture to derive 20-DoF joint-angle labels from three-dimensional marker trajectories.

### Precise Hand–Arm Teleoperation System via Markerless Vision and High-Density sEMG Fusion

Dalian University of Technology

2024.09 – 2025.07

- Developed a markerless vision-based hand tracking method to estimate global hand position and angles from an RGB-D camera, validated against Vicon optical motion capture ground truth.
- Proposed a dual-input CNN fusion model that combines markerless hand keypoints and high-density sEMG signals for hand gesture recognition, achieving higher accuracy than single-modality baselines.
- Designed a hybrid finger-force regression network based on 1D-CNN and Bi-LSTM to estimate finger interaction forces during grasping tasks, enabling adaptive manipulation of fragile objects.
- Built a wireless hand-arm teleoperation system in ROS2, enabling real-time mapping of hand posture, gestures, and grasp force to a robotic manipulator through multimodal RGB-D and EMG inputs.

### A Distributed High-Density sEMG-driven Soft Glove System for Stroke Rehabilitation

Dalian University of Technology (*Bachelor graduation thesis*)

2023.10 – 2024.07

- Developed a pneumatic soft rehabilitation glove system using Arduino, enabling independent finger flexion and abduction.
- Designed a high-density sEMG acquisition sleeve and optimized electrode placement based on sEMG energy distribution mapping and impedance analysis.
- Implemented and evaluated multiple machine learning and deep learning models for EMG-based hand gesture classification, ultimately adopting a PSO-optimized neural network to classify five hand gestures.

### Industrial & Mechanical Structure Design

Design Director, Venture Electronics, Shenyang, China (*High Definitive Audio Amplifier and DAC*)

2021.09 – present

Product Manager, Dalian University of Technology (*Wireless High-Density sEMG acquisition system*)

2023.11 – present

## SKILLS

Languages: Mandarin (Native); English (CET6: 535)

Programming: MATLAB, Python, PyTorch, PyTorch Lightning, ROS2, Mujoco

Design Software: Curved surface modeling (Rhino), Engineering modeling (Inventor), Rendering (KeyShot)

Interests: Soccer, HiFi, Singing, Billiards, Coffee